MAINGAU

BUILD DAYS CHALLENGE

One customer, every channel

Designing a Next Best Action engine for MAINGAU Energie

NBA / NBO

Kick-off · Day 1

“

If we have 4.5 million customers, we shouldn’t have one store. We should have 4.5 million stores.

Jeff Bezos, Amazon, 1998

71%

of consumers expect personalised interactions

76%

get frustrated when it doesn’t happen

Sources: Washington Post, Nov 1998 · McKinsey, Next in Personalization 2021

ABOUT MAINGAU

Energy goes tech

An energy and tech company from the Rhein-Main region, and one of the fastest-growing energy providers in Germany.

€1.3 bn

Revenue in 2025

1 M+

Customer relationships across Europe

~350

Employees in the Rhein-Main region

1907

Founded, 115+ years of experience

OUR CORE VALUES

Ambition

Technology

Innovation

OUR PORTFOLIO

Five businesses, built up since 1907



Since 1907

Energy supply

Gas since 1907, electricity since 2007, incl. heat-pump electricity, across Germany

~500k

Customers in Germany



Since 2018

Autostrom

EV charging via app, card or chip, plus THG bonus and car sharing

927,000+

Charge points in Europe



Since 2021

Mobile

Mobile phone contracts with flexible data plans

~27k

Customers



Since 2016

Online shop

Smart home devices, wallboxes and brand products

~43k

Orders per year



New

Smart Living

Smart meter and smart home today; Smart Charging and HEMS coming soon

PoC

MVP go live End of October 2026

OUR PORTFOLIO

One customer base, many systems

All lines	Energy	Mobile	Online shop	Smart Living	Autostrom	Autostrom	Marketing
CRM	**Billing ERP**	**Telco backend**	**Shopware 6**	**HEMS cloud**	**Autostrom backend**	**Roaming hub**	**Newsletter tool**

IN PROGRESS

Unified master view on Databricks

One customer_id links every source

MISSING

A decision layer on top

Which offer, which channel, which tone, and when

FOR THE BUILD DAYS

Integration is given: you get one consistent mock dataset with a single customer_id. **Build the decision layer, not the pipes.**

THE CHALLENGE

Build a clickable prototype that recommends the right offer, channel and tone for each customer, and a marketing dashboard to steer it.

Best means: the highest expected gain in CLV.

TODAY

Campaigns reach customers in broad, product-based segments with static criteria. Solid, but not personal, not dynamic, and blind to customer lifetime value (CLV).

FOR EXAMPLE

A customer deregisters their gas meter – a clear sign of a new heat pump – but gets no offer for heat-pump electricity.

FOR EXAMPLE

A customer buys a wallbox in our shop, but gets no offer for charging at home or on the road with Autostrom.

FOR EXAMPLE

A customer clicks the heat pump story in three newsletters in a row. Nobody follows up – only an unsubscribe triggers a reaction.

FOR EXAMPLE

A loyal high-value customer gets the same action, at the same time, in the same tone as someone who joined last week.

THE CHALLENGE

What “best” means: the highest expected gain in CLV

Expected value of an action = p(accept) × (CLV after – CLV now) – contact cost

RAISES CLV

Cross-sell and upsell: new margin, and bundled customers churn less

Retention: a renewed price guarantee avoids a cancellation

Solving a service case: removes the churn risk it created

LOWERS CLV

Service cases: handling costs and higher churn – open complaints, failed charging, dunning

A sales offer during an open service case: it annoys, churn goes up

Every contact has a cost, and too many contacts cause fatigue

EXAMPLE

An Autostrom driver with three failed charging sessions: a home electricity offer scores negative, a proactive service fix scores clearly positive. **The engine picks the fix.**

CLV, margins, churn drivers and all parameters are precomputed in the mock data (customer_value, value_assumptions).

THE CHALLENGE

Three questions to answer

01

Decide

Which offer, channel and tone (Du/Sie) does the engine pick per customer, ranked by expected CLV gain?

Respects consent, exclusions and contact limits

02

Explain

Can a marketer see in one line why the engine chose this action – and change it?

Inputs · rules · scores · chosen action, all logged

03

Learn

How would reactions improve the engine over time, and how do you prove an action caused a result?

Concept is enough · control group · uplift

THE CHALLENGE

Reference pipeline

1
Trigger
Event or scheduled run selects a customer

2
Eligibility
Consent, products held, exclusions, frequency

3
Scoring
Expected change in the customer's CLV – can be negative

4
Arbitration
Rank by expected CLV gain, drop negative actions, respect fatigue and limits

5
Channel & tone
Where and how to address the customer

6
Content
e.g. LLM, within approved templates

7
Delivery & log
Track response, back to step 1

EXAMPLE

Electricity customer writes formal emails, never opens newsletters, logs into the portal weekly.

Engine skips the newsletter and shows a portal pop-up, addressed with "Sie".

DAY ONE

Start with rules, the precomputed CLV and acceptance rates from the action history. ML is optional – log every decision and keep a control group.

KEY AIM

One clickable prototype, three parts

ENGINE

Recommendations

Per customer: offer, channel, tone

Expected CLV gain and a one-line reason

Runs on the mock data

FOR MARKETING

Orchestration dashboard

Top NBAs across customers and segments

Set offers, priorities and contact limits

Approve content, channels and tone

FOR CUSTOMER SERVICE

Customer 360° dashboard

Current status: products, contracts, open cases

CLV at a glance

Most favourable NBAs, ranked, with a reason

PLUS

One slide on target architecture: how this would run on Databricks with MCP data access.

THE CHALLENGE

Deliverables and judging

CORE – EVERY TEAM

1 Clickable prototype
Offer, channel, tone, expected CLV gain and reason per customer

2 Marketing dashboard
Steer, approve and monitor NBAs

3 Customer 360° dashboard
Status, CLV and top NBAs for service agents

4 Architecture concept
One slide: Databricks, MCP, how it would run

5 Pitch
8 minutes plus 4 minutes Q&A, in English

DEEP DIVE – PICK ONE

A Feedback loop & uplift

B Data freshness & event replay

C Content generation

D Tone detection

JURY WEIGHTING

Technical quality	25%
Feasibility	25%
Impact for MAINGAU	25%
Innovation	15%
Pitch	10%

Checked on reference customers hidden in the data:

· expected CLV gain of the recommended actions

· zero consent or exclusion violations

· a clear reason for every decision

THE CHALLENGE

Mock data and hard rules

750 Synthetic customers

16 Tables, one customer_id

640 Live events to replay

10% Control group in action history

Consent is binding
No channel without consent; withdrawal stops actions at once

Profiling needs consent
Personalisation only for customers who agreed to profiling

Explainability
Every decision logged: inputs, rules, scores, chosen action

Exclusions beat offers
Dunning, open complaints or cancellations block sales content

No manipulation
No personalised prices, no fake urgency, no targeting of customers in hardship

Mock data only
No real customer data, no scraping of MAINGAU systems

Human in the loop
The engine proposes; marketing approves. No autonomous sending

Payment data only as exclusion
Never used to score creditworthiness

Language
Pitch in English; customer content in German

EU AI Act applicability will be checked before any production use.

MAINGAU Energie

Let’s build

A clear, compliant recommendation for every customer – one that marketing can steer and that pays off in customer value.

MOCK DATA & DATA DICTIONARY: SHARED ON [DATUM]

YOUR CONTACTS

Caro
Team Lead Marketing

Raphael
Data Architect

Jan
Product Manager B2C Energy

Felix
Digital Channels & E-Commerce